

INT 403 | INT531

Site Reliability Engineering

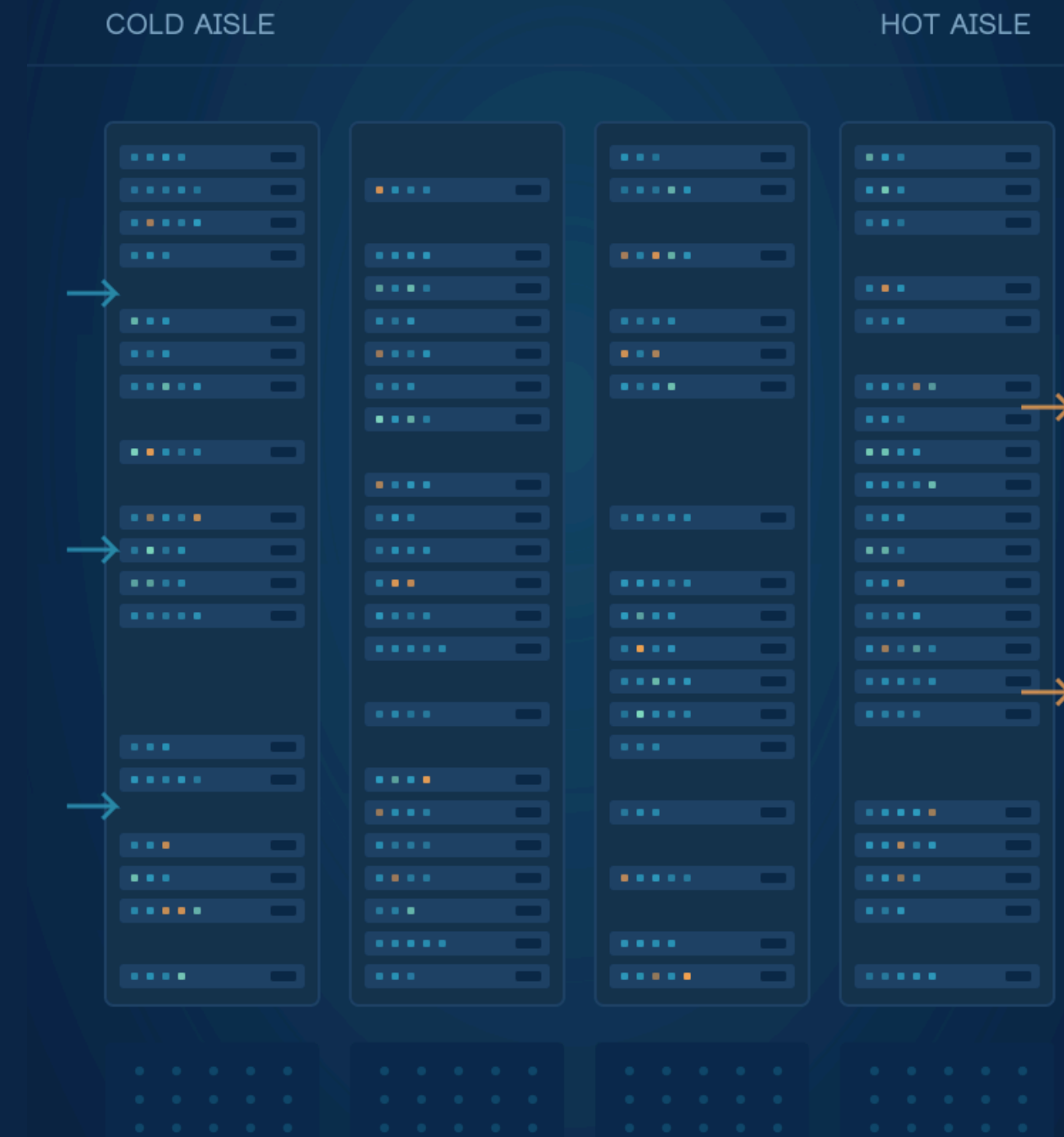
Designing, measuring and running systems that stay up

Week 1 · Introduction to SRE and the idea of reliability



School of Information Technology, King Mongkut's University of Technology Thonburi

Lecture 2 h + Lab 2 h | Content updated 2026



What classes did we take last semester



What classes did we take last semester



What we cover today

Lecture 2 h + Lab 2 h

01

Why SRE exists

A real outage, its blast radius, and what failure costs

02

What SRE actually is

Definition, principles, and how it differs from DevOps and classic ops

03

How reliability gets measured

The price of each nine, and why 100% is the wrong target

04

Toil and the work that should disappear

The six tests, and the 50% ceiling on operational work

05

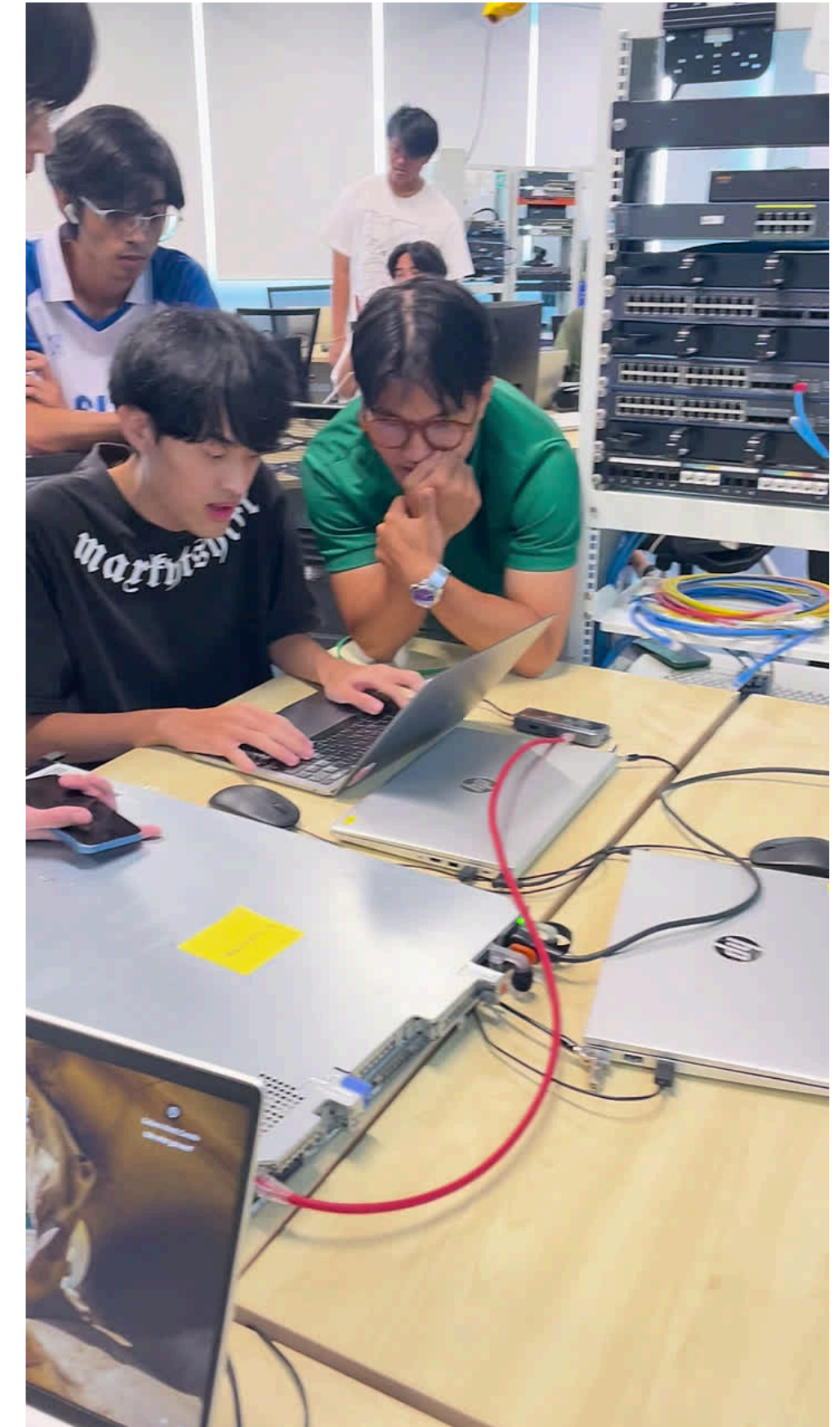
SRE in 2026 and the FDE career

Five trends, plus the Forward Deployed Engineer role

06

Lab 1 — moving racks into containment

Planning real work in the data centre against ISO/IEC 27001



Session Learning Outcomes

By the end of this session you will be able to...

SLO 1.1 Explain where SRE came from and why it emerged at large-system scale

CLO1

SLO 1.2 Compare SRE, DevOps, Platform Engineering and classic system administration

CLO1

SLO 1.3 Define toil, sort work into toil or engineering, and explain the 50% ceiling

CLO1

SLO 1.4 Explain in economic terms why 100% reliability is the wrong target

CLO1

SLO 1.5 Produce an infrastructure change plan that references a recognised security standard

CLO3, CLO7



01

Why SRE exists

Reliability is never free — we start with an outage the whole internet felt

Case study: the AWS us-east-1 outage of 20 October 2025

One small defect turned into a global event

~15 h

Time until every service was fully recovered

140+

AWS services affected

1 bug

A race condition in automated DNS management

~\$581M

Estimated insured losses

How it unfolded

23:48

Automated DNS management for DynamoDB overwrote itself, wiping the DNS records for the service endpoints

00:38

Engineers identified the DNS fault, but the automation could not repair itself — records had to be fixed by hand

02:25

DynamoDB came back, but services that depended on it kept failing in a chain

14:20

Everything recovered — the time spent recovering was several times the time spent fixing the root cause

Sources: AWS post-event summary; ThousandEyes, AWS Outage Analysis (Oct 2025); CyberCube (loss estimate)

Four lessons that come back all term

Each one is the subject of a later week

Weeks 3 · 12

Dependencies fail in chains

AWS's own internal services depended on DynamoDB, so the tools needed to fix the outage were broken too — this is cascading failure and control-plane dependency

Week 7

Automation that cannot repair itself

Automation without safety guards does damage faster than a human can. The team had to disable it and repair records manually

Week 15

Recovery is harder than the original failure

The root cause was fixed in about 3 hours; it took another 12 to drain backlogs and bring everything back

Week 16

Transparency builds trust

AWS published a detailed public analysis, so the whole industry learned from it — that is a blameless postmortem

So what about 2026? The pressure is rising

Data and forecasts as of mid-2026

Outages recorded in H1 2026

30,246

Across 1,082 providers spanning cloud, SaaS, AI, payments and observability tooling

The 2026 forecast

2 events

Forrester expects at least two multi-day hyperscaler outages this year, as providers shift spending toward AI data centres and away from maintaining older infrastructure

Developers using AI daily

90%

DORA 2025 found AI does raise delivery throughput — and raises delivery instability along with it

Failure modes that only showed up in 2026

- Physical and geopolitical risk — AWS reported that data centres in the Middle East took physical damage from regional conflict
- AI providers became production dependencies — one LLM outage disrupted education platforms, developer tooling and customer support at once
- The provider's own automation as the cause — an automated account suspension left one customer with almost nothing running
- Cloud concentration risk is now a regulatory question, not just a technical one

Sources: IncidentHub, H1 2026 Cloud & SaaS Reliability Report; Forrester Predictions 2026: Cloud Computing; DORA 2025 State of AI-assisted Software Development



02

What SRE actually is

From operations done by hand to an engineering problem solved with code and data

Definition and origin

Google, 2003 — Ben Treynor Sloss

Ben Treynor Sloss described **SRE** as what you get when software engineers design and run the operations function, instead of hiring more system administrators as the system grows

The problem to solve

- The system grows tenfold; the headcount cannot
- Manual operations work scales linearly with load
- Development and operations teams are pulled toward opposing goals

How SRE thinks

- Treat running systems as a software engineering problem
- Decide with numbers, not with instinct or seniority
- Accept risk at a level worked out in advance

What good looks like

- Operational load grows more slowly than the system
- Developers and operators share one target
- A team that can sustain the work without burning out on call

How SRE differs from DevOps and classic system administration

None of these replaces the others — they answer different questions

	Traditional Ops	DevOps	SRE	Platform Engineering
Core question	Is the system still up?	How do we ship faster and more often?	Is the system reliable enough, and how do we know?	How do we let other teams do this themselves?
What it is	A culture and a set of practices	A culture and a set of practices	A job title and a real role	A job title and an internal product team
Headline metric	Uptime and ticket counts	The four DORA metrics	SLOs and error budgets	Developer Experience, Adoption
What gets delivered	Firefighting fixes	Pipelines and shared culture	Code, automation and SLOs	Internal Developer Platform

Aside: “class SRE implements DevOps” — SRE is one concrete, measurable way to actually do DevOps

The seven principles of SRE

Each one gets unpacked across the fifteen weeks

Week 2

Embracing Risk

Take risk at a level you calculated, instead of chasing perfection

Week 2

Service Level Objectives

Set targets that are measurable from the user's side, not the server's

Week 7

Eliminating Toil

Repetitive work gets replaced by code, not by more people

Weeks 4–5

Monitoring & Observability

If you cannot explain what the system is doing, you cannot run it

Week 7

Automation

Automation always needs limits and an emergency stop

Week 10

Release Engineering

Releases must be repeatable, verifiable and reversible

Week 12

Simplicity

Unnecessary complexity is reliability debt

The hard line: the 50% rule

An SRE team spends no more than half its time on toil; the rest must be engineering that leaves less toil next month

The 50% ceiling is the hardest one in practice, because organisations tend to measure infrastructure teams by tickets closed rather than tickets prevented



03

How do you measure reliability?

Before you can improve anything, everyone has to agree what “good enough” means

What each nine costs you

Downtime you can spend and still meet the target (per 30 days)

99% 7.2 hours Internal, non-critical systems	99.9% 43.2 minutes General-purpose services	99.95% 21.6 minutes Systems with external users	99.99% 4.3 minutes Payments and transactions	99.999% 26 seconds Telecom networks
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Three things to take from this table

- Each extra nine cuts allowed downtime tenfold — but usually costs far more than tenfold
- Components in series are always less reliable than the weakest one: three 99.9% services chained give roughly 99.7%
- Users do not feel the number, they feel when it broke, for how long, and who it hit — 43 minutes at 3 a.m. is not 43 minutes on registration day

Why 100% is the wrong target

Examples from systems the students already use

Fact one

Users never experience 100% anyway

Between the user and our system sit a phone, Wi-Fi, a mobile carrier and DNS. If that path is 99.6% reliable, going from 99.99% to 99.999% is invisible to the user

Fact two

The last nine costs you the features you did not build

Budget, time and people are finite. Every hour spent chasing perfection is an hour not spent solving another user problem. Reliability is a business decision, not a point of technical pride

Different targets inside one organisation

Course registration

99.95%

Ten minutes down on opening day means thousands of students lose their slot — this needs a fallback and a rehearsal every term

LMS and assignment submission

99.9%

Night-time outages hurt most, because that is when work actually gets submitted — the SLO should vary by time of day

Faculty public website

99%

Half a day down bothers almost nobody — multi-region is not worth paying for

What toil is, and why it has to go

Being busy is not the same as being useful

Toil is work needed to keep the system alive that leaves it no better than before

The six tests — the more that apply, the clearer the toil

- 1 ทำด้วยมือ (Manual)
- 2 ทำซ้ำ ๆ (Repetitive)
- 3 เขียนโปรแกรมแทนได้ (Automatable)
- 4 แก้เฉพาะหน้า (Tactical)
- 5 ไม่เหลือคุณค่าถาวร (No enduring value)
- 6 โตตามขนาดระบบ (Scales linearly)

Items three and four are exactly what today's lab puts in front of you

Toil we actually have in the school's data centre

- Resetting user passwords one at a time in an admin console
- Restarting the same stuck service every Monday morning
- Walking the racks writing serial numbers on paper, then retyping them into a spreadsheet
- Plugging a console cable into each box to check its firmware version
- Copying PUE readings off the UPS display into a monthly report

So what is not toil?

Work that needs judgement and leaves lasting value — architecture, building automation, analysing a postmortem. Just as tiring, but you only do it once



04

SRE in 2026 and where the careers are

What changed, what never changes, and the role the market is now asking for

Five trends shaping SRE work this year

And what each one means for what you should practise

1	AI amplifies; it does not fix	DORA 2025 พบว่า AI เพิ่มความเร็วในการส่งมอบ แต่เพิ่มความไม่เสถียรด้วย ทีมที่มีพื้นฐานดีจะยิ่งดี ทีมที่พื้นฐานแย่จะยิ่งพัง	ฝึก: SLO และ Rollback ต้องแน่นก่อนเร่งความเร็ว
2	AI providers are the new dependency	LLM API กลายเป็นส่วนหนึ่งของ Production Path การล่มของผู้ให้บริการรายเดียวจึงกระทบเป็นวงกว้าง	ฝึก: Timeout, Fallback และการออกแบบให้ลดระดับบริการได้
3	Platform Engineering	องค์กรสร้างแพลตฟอร์มภายในเพื่อให้ทีมพัฒนา Deploy เองได้อย่างปลอดภัย SRE ขยับไปเป็นผู้สร้าง Golden Path	ฝึก: มองระบบภายในเป็นผลิตภัณฑ์ที่มีผู้ใช้
4	OpenTelemetry as the common standard	การเก็บ Metrics, Logs, Traces ย้ายมาอยู่บนมาตรฐานเดียว ลดการผูกติดกับผู้ขายรายใดรายหนึ่ง	ฝึก: Instrumentation ด้วย OTel ตั้งแต่สัปดาห์ที่ 5
5	Cost and power are real constraints	การแข่งขันสร้างศูนย์ข้อมูล AI ทำให้ไฟฟ้า ความเย็น และพื้นที่ Rack กลายเป็นทรัพยากรที่ต้องบริหารอย่างจริงจัง	ฝึก: FinOps, PUE และการจัดการ Containment

Unchanged since 2003: find out where the user hurts, measure it, then fix it

The role everyone is hiring for: Forward Deployed Engineer

An engineer who embeds with the customer and owns the system from day one to the day it breaks



บทบาทนี้ถูกริเริ่มที่ Palantir ตั้งแต่ราวปี 2005 และในปี 2026 ถูกนำมาใช้อย่างกว้างขวางโดย OpenAI, Anthropic, Google Cloud, Databricks, Salesforce และ Scale AI · เส้นแบ่งจากงานที่ปรึกษาคือ ที่ปรึกษาส่งมอบรายงาน แต่ FDE ส่งมอบระบบที่ทำงานอยู่จริง

How FDE differs from adjacent roles, and how to prepare

Job-market data as of mid-2026

~800%

การเติบโตของประกาศรับสมัคร FDE

2005

ปีที่ Palantir ริเริ่มบทบาทนี้

\$160–280K

ฐานเงินเดือนระดับกลางที่ OpenAI (SF)

~50%

สัดส่วนเวลาที่อยู่หน้างานลูกค้า

	ที่ปรึกษา	Solutions Engineer	FDE	SRE
ส่งมอบอะไร	รายงานและข้อเสนอแนะ	ตั้งค่าผลิตภัณฑ์ที่มีอยู่ให้ลูกค้า	โค้ดและระบบที่รันจริงในสภาพแวดล้อมลูกค้า	ระบบอัตโนมัติ SLO และเครื่องมือที่ทำให้ระบบอยู่รอด
ใครถูกเรียกตอนพัง	ไม่ถูกเรียก งานจบเมื่อส่งรายงาน	ส่งต่อให้ทีม Support	คนเดิมที่วางแผนวันแรก ถูกเรียกในเดือนที่หก	ทีมที่ถือ Error Budget ของบริการนั้น

เตรียมตัวอย่างไรตั้งแต่ยังเรียนอยู่

ส่งของขึ้น Production จริง

ไม่ใช่แค่ Demo — ต้องมีระบบที่มีคนใช้จริง และมีคนถูกเรียกเมื่อมันพัง

ฝึกแก้ปัญหาที่กำกวม

สัมภาษณ์ FDE เน้นการถามให้ชัดก่อนลงมือมากกว่าการแก้อัลกอริทึม

สื่อสารกับคนที่ไม่ใช่วิศวกร

ทักษะนี้ถูกวัดเท่ากับการเขียนโค้ดในกระบวนการคัดเลือก

รู้เรื่องความน่าเชื่อถือ

SLO, Observability, Rollback — คือสิ่งที่วิชานี้สอนทั้งหมด

Sources: MarkTechPost (May 2026); Paraform (Jun 2026); OpenAI and Palantir job listings — pay figures are point-in-time and vary by level and location



Lab 1 — planning the move into the new containment racks

Real work in the school's data center · framed against ISO/IEC 27001:2022

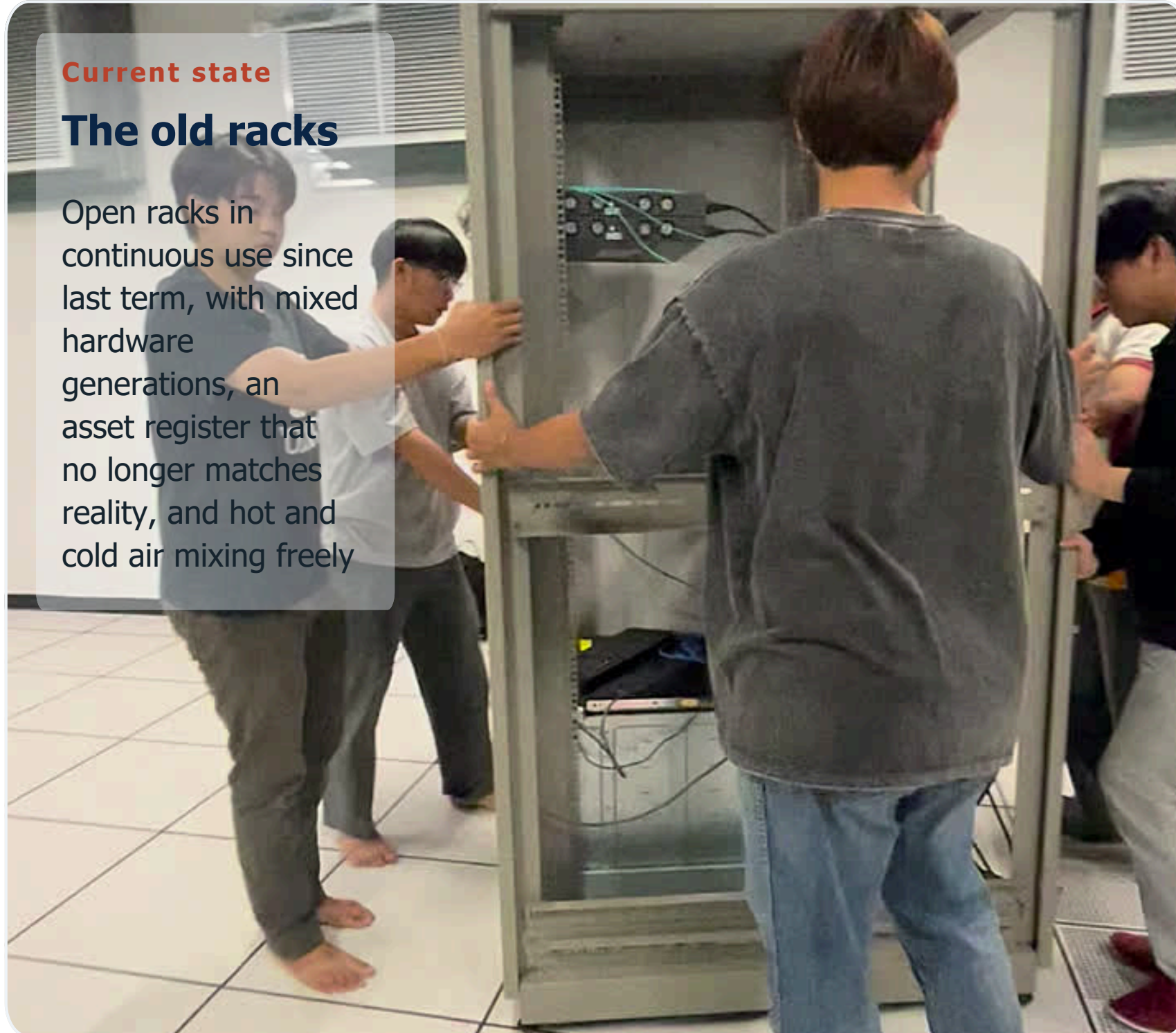
The problem: move everything from the old racks into new containment

Where things stand at the start of term

Current state

The old racks

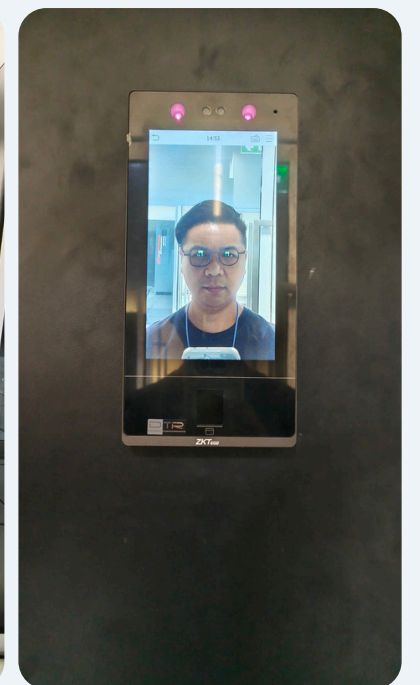
Open racks in continuous use since last term, with mixed hardware generations, an asset register that no longer matches reality, and hot and cold air mixing freely



Target state

The new containment racks

Racks with hot and cold aisle containment, which stops the two air streams mixing, improves PUE and supports higher power density per rack



The problem: move everything from the old racks into new containment

Where things stand at the start of term

Current state

The old racks

Open racks in continuous use since last term, with mixed hardware generations, an asset register that no longer matches reality, and hot and cold air mixing freely

Target state

The new containment racks

Racks with hot and cold aisle containment, which stops the two air streams mixing, improves PUE and supports higher power density per rack

Constraints

What limits us

Some services cannot stop during working hours, so the work happens out of hours, inside a fixed window, and must be reversible

Why this is SRE work, not just lifting equipment

Change Management

Every change needs a plan, an approver and a named owner

Blast Radius

If you pull the wrong switch, how many services go dark and who feels it?

Rollback Plan

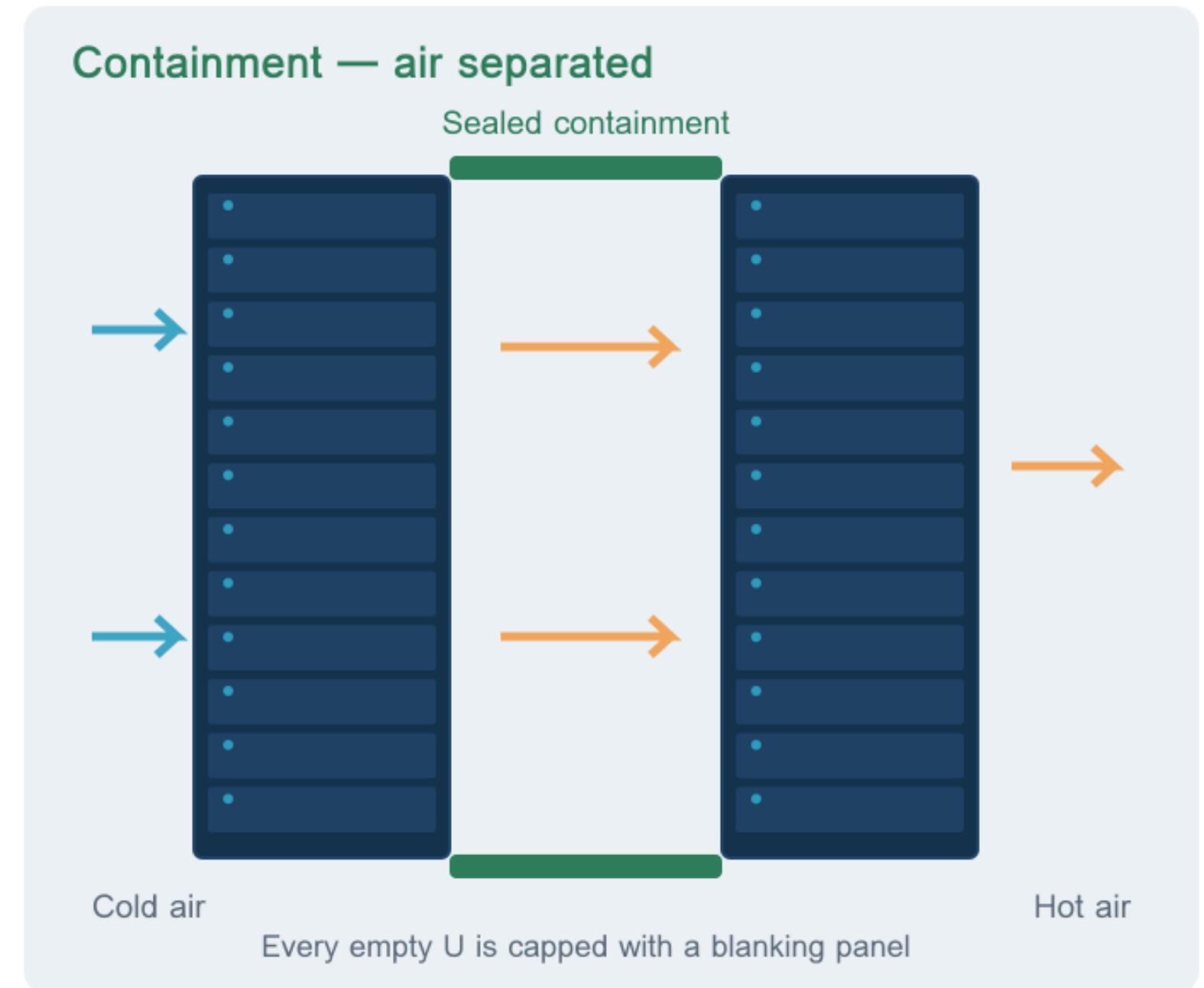
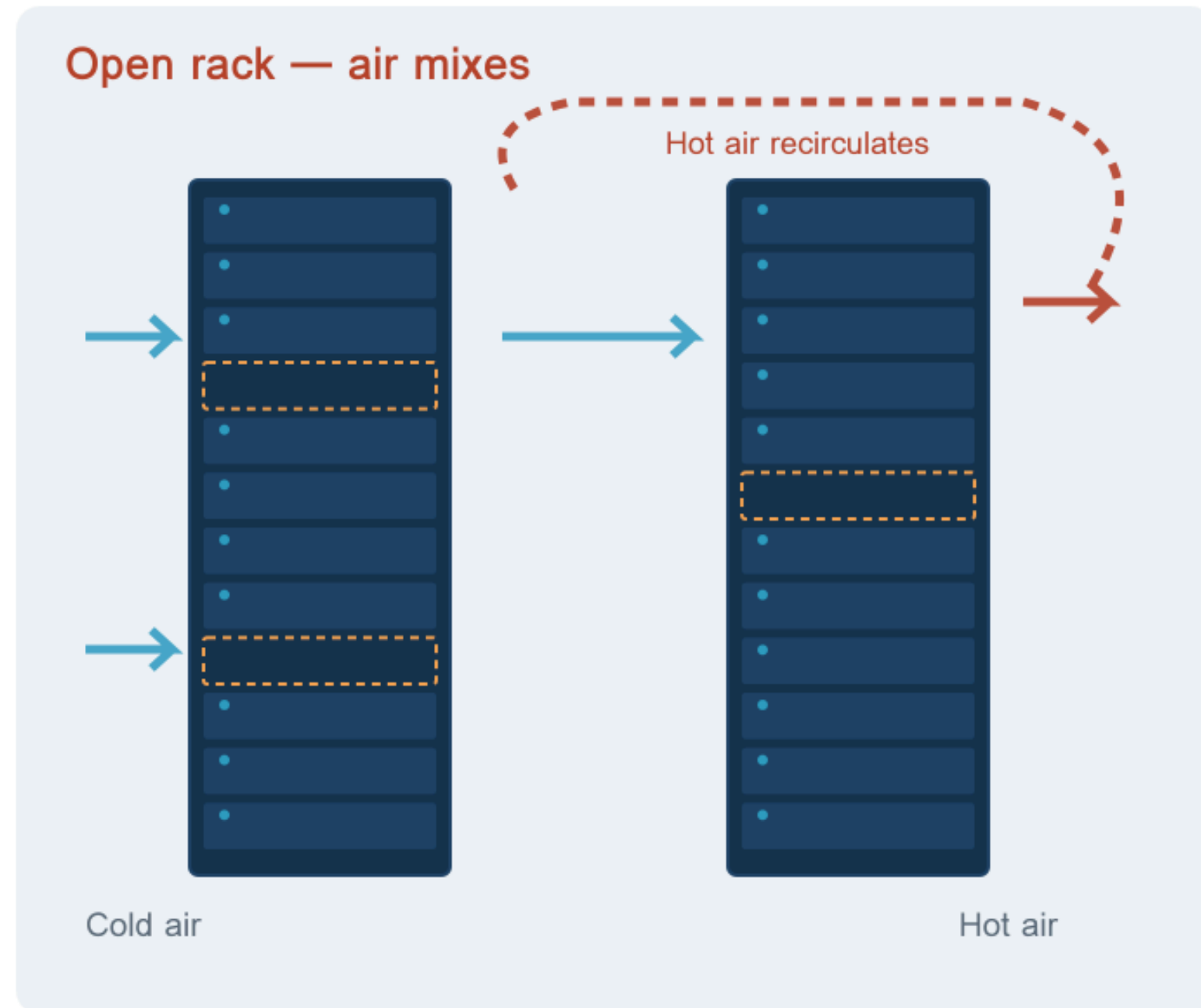
If a box does not boot in the new rack, how many minutes to get back to where we started?

Asset Inventory

If you do not know what is in the rack, you cannot assess risk or recover

Why containment, specifically

A problem you cannot fix by adding more cooling — only by separating the air paths



1 Hot air recirculating to the front raises intake temperature above design, so the cooling plant works harder for no benefit

2 Every open, uncapped U is a shortcut for hot air — blanking panels are not decoration

3 The measurable result is a better PUE, and higher power density per rack in the same floor space

The control framework: ISO/IEC 27001:2022 Annex A

Every lab activity must map to at least one control

A.5.9	Inventory of information and associated assets	Build an inventory of every device in the old racks, naming the asset owner and the services involved
A.5.37	Documented operating procedures	Write a step-by-step method of procedure that someone outside the team can follow
A.7.4	Physical security monitoring	Log entry and exit from the data centre, and control any contractors working on site
A.7.8 / A.7.12	Equipment siting and protection, and cabling security	Lay out the new rack: U positions, cable routes, and separation of power from signal
A.7.10 / A.7.14	Storage media, and secure disposal or re-use	Handle the disks in any device being retired — data must be erased securely, with evidence
A.7.11	Supporting utilities	Check load per phase, UPS capacity and cooling before anything moves into the new rack
A.8.32	Change management	Raise a change request stating impact, rollback plan and the approving authority

Note: control numbers follow ISO/IEC 27001:2022 Annex A. Check them against the edition your organisation actually applies before submitting

The work in five phases

Phases 1–2 finish in this session · phases 3–5 continue outside class

1 Survey and inventory

Walk the old rack one U at a time, recording make, model, serial, U position, power draw, ports in use and the services running

Output: inventory sheet

A.5.9

2 Analyse dependencies and risk

Map what connects to what, which services die if a device is pulled, and rate how critical each item is

Output: dependency map + risk register

A.8.32

3 Design the new rack layout

Place each device in the new containment rack, accounting for airflow, load per phase, weight and cable length

Output: rack elevation diagram

A.7.8 / A.7.11

4 Write the procedure and the rollback plan

Produce a step-by-step MOP with time estimates, owners, verification points and abort criteria

Output: MOP + rollback plan

A.5.37

5 Rehearse and review

Walk another group through the plan on paper, find where it breaks, and fix it before anyone touches hardware

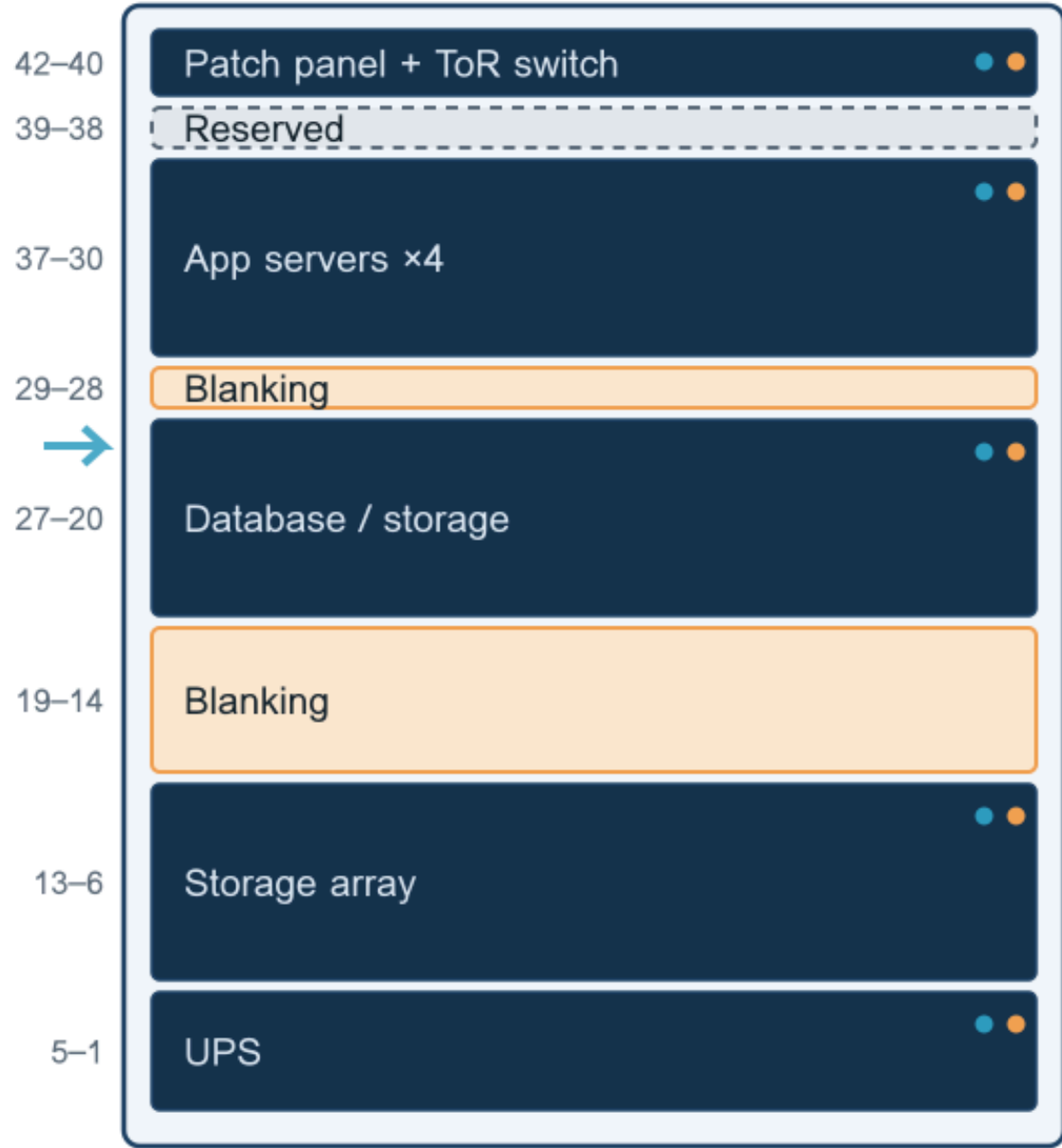
Output: review log + revised plan

A.5.37

A worked phase 3 output: the rack elevation

This format is the minimum standard for submission — not just a list of equipment

Rack elevation — worked example



- Active equipment
- Blanking panel (mandatory)
- Reserved for growth

- 1 Heavy gear (UPS, storage) sits low — lowers the centre of gravity
- 2 Every empty U is capped so hot air cannot recirculate
- 3 Dual-PSU devices are split across PDU A and PDU B
- 4 Load per phase kept under 80% of rating



What you submit, and the safety rules

15-20 Groups | Groups of 3-4 • due at the start of week 2

Lab 1 marking (10 marks)

- Inventory completeness — every item has a serial, U position, owner and associated service (3 marks)
- Quality of the dependency map — shows what dies when something is pulled, with criticality ratings (2 marks)
- Clarity of the MOP — someone outside the team can follow it, with time estimates and verification points (2 marks)
- A rollback plan that would actually work — abort criteria and time to return to the previous state (2 marks)
- Correct, well-reasoned mapping to ISO/IEC 27001 controls (1 mark)

Safety rules — no exceptions

- Do not unplug, disconnect or power down anything during this session
- Enter the data centre only with staff or the instructor present, and sign in and out every time
- Do not photograph asset tags, credentials or network diagrams and share them outside the course
- Report anything abnormal — burning smell, unusual noise, damaged cabling — immediately, and do not fix it yourself

Team roles — you rotate through them next week, and they become the incident response roles in week 15

Change Owner

Owns the change plan and decides whether to proceed or abort

Inventory Lead

Accountable for the accuracy of every line in the asset register

Safety Officer

Owns physical safety and compliance with data centre rules

Scribe

Records times, who did what, and what was found, for later verification

Assignments and how to prepare for next week

Week 2: SLI / SLO / SLA and error budgets

Before the next session

- Read Google SRE Book chapter 4 (Service Level Objectives)
- Complete pre-class quiz 1 on the LMS
- Pick one service you personally use every day, for assignment 1

Lab 1 group work

- Finish the inventory sheet and the dependency map
- Push both to your group repository along with the data centre access log
- Bring your unresolved questions to the next session

Have your environment ready

- The course sample application running under Docker Compose
- A baseline response-time measurement written down

The question that opens next session: if you had to tell the university president in a single number whether the registration system is good enough, what number would you use?

References and further reading

Grouped by how much you need them for this session

Core texts

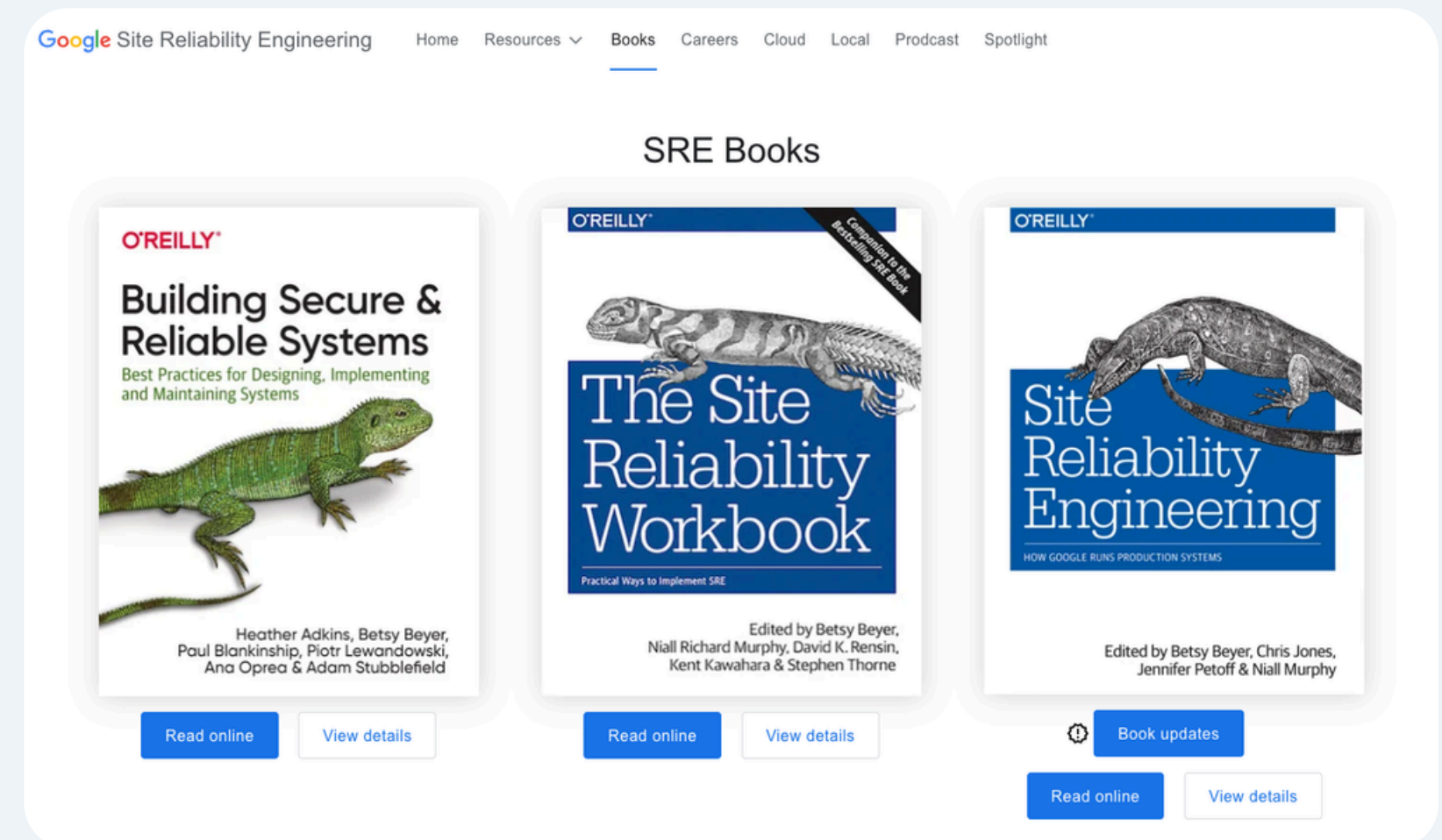
- Beyer, B., Jones, C., Petoff, J., & Murphy, N. R. (2016). **Site Reliability Engineering**. O'Reilly. (ch. 1–3, 5) — free at sre.google/books
- Beyer, B. et al. (2018). **The Site Reliability Workbook**. O'Reilly. (ch. 1–2)

Case studies and 2025–2026 data

- AWS. **Summary of the Amazon DynamoDB Service Disruption in Northern Virginia (us-east-1)**, October 2025 — aws.amazon.com/message/101925/
- ThousandEyes. **AWS Outage Analysis**: October 20, 2025.
- IncidentHub. **H1 2026 Cloud and SaaS Reliability Report** (July 2026).
- Forrester. **Predictions 2026**: Cloud Computing.
- InfoQ. **Coinbase Postmortem on a Localized AWS Failure** (June 2026).

Standards and lab references

- **ISO/IEC 27001:2022** — Information security management systems (Annex A controls)
- **ISO/IEC 27002:2022** — implementation guidance for the controls
- **ISO/IEC 22237 series** — Data centre facilities and infrastructures
- **ISO/IEC 30134-2** — Power Usage Effectiveness (PUE)
- **DORA**. 2025 State of AI-assisted Software Development. dora.dev



Today in three sentences

01

Reliability is a product attribute you design, measure and pay for — it does not happen on its own

02

The goal is not a system that never fails, but one that fails predictably and comes back before users give up

03

Repetitive manual work is a debt you pay every month — half the team's time must go into making it disappear

Week 2 · SLI / SLO / SLA and error budgets — we start putting numbers on everything we said today

A note on the artwork (instructor slide)

Not for projection — use this when swapping or editing the images in this deck

Artwork in this deck

The server-room image on the title slide, the containment diagram and the rack elevation were drawn as vectors specifically for this course, so they can be recoloured, edited and reused with no third-party licensing conditions.

If you want real photographs instead

Free sources cleared for commercial use: Unsplash, Pexels, Pixabay · Search terms that work well: data center server rack, hot aisle containment, network operations center, structured cabling, rack elevation

A caution under control A.7.4

Do not put photographs of the actual data centre showing asset tags, network diagrams or screens with credentials into any deck shared outside the course.